

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning from data, understanding language, recognizing images, solving problems, making decisions, and interacting with users. AI systems use algorithms and data to analyze information and improve their performance over time.

Examples of AI include virtual assistants, recommendation systems, chatbots, self-driving vehicles, and medical diagnosis tools.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence (AI), Machine Learning (ML), Generative AI, and Agentic AI are related technologies, but they differ in their capabilities and applications.

Artificial Intelligence (AI)

Artificial Intelligence is the broad field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include reasoning, problem-solving, decision-making, language understanding, and pattern recognition.

Example: A chess-playing computer program.

Machine Learning (ML)

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed for every task. ML algorithms identify patterns in data and make predictions or decisions based on those patterns.

Example: Email spam detection systems.

Generative AI

Generative AI is a specialized branch of AI that creates new content such as text, images, videos, audio, and computer code. It learns from large datasets and generates outputs that resemble human-created content.

Example: ChatGPT generating essays or answers to questions.

Agentic AI

Agentic AI refers to AI systems that can autonomously plan, make decisions, and take actions to achieve specific goals. These systems can perform multi-step tasks with minimal human intervention.

Example: An AI assistant that schedules meetings, sends emails, and manages tasks automatically.

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

1. Llama 3

Llama 3 is an open-source large language model developed by Meta. It is designed for tasks such as text generation, summarization, question answering, and coding assistance. The model offers strong performance while being accessible for research and development purposes.

2. Mistral 7B

Mistral 7B is a lightweight yet powerful language model developed by Mistral AI. It is known for its efficiency and strong performance despite having fewer parameters than many larger models. It is widely used for chatbots, content generation, and natural language processing tasks.

3. Falcon 7B

Falcon 7B is an open-source language model developed by the Technology Innovation Institute (TII). It is trained on large datasets and can perform text generation, summarization, translation, and conversational AI tasks. Falcon models are popular because of their strong open-source capabilities.

Q4. What Is the Difference Between Traditional Programming and AI-Assisted Development?

Traditional Programming and AI-Assisted Development are two different approaches to software development. While traditional programming relies entirely on human-written instructions, AI-assisted development uses artificial intelligence tools to support developers during the coding process.

Traditional Programming

Traditional programming involves writing explicit instructions that a computer follows to perform a task. Developers must design the logic, write the code, test the program, and fix errors manually. The computer only executes the rules provided by the programmer and cannot learn or adapt on its own.

Characteristics of Traditional Programming

Requires manual coding of all program logic.

Follows predefined rules and instructions.

Debugging and testing are mainly performed by developers.

Limited automation in software development.

Development can be time-consuming for complex projects.

Example: A programmer writes every line of code required for a calculator application.

AI-Assisted Development

AI-assisted development uses artificial intelligence tools to help developers write, understand, and improve code. These tools can generate code snippets, suggest solutions, detect errors, create documentation, and assist in debugging. AI acts as a helper, increasing productivity and reducing development time.

Characteristics of AI-Assisted Development

AI can generate code suggestions automatically.

Helps identify and fix coding errors.

Assists in writing documentation and explanations.

Increases development speed and efficiency.

Developers still review and validate AI-generated outputs.

Example: An AI coding assistant generates a function based on a developer's description of the task.

Q5. Explain AI Hallucinations with One Example

AI hallucination occurs when an AI system generates information that appears believable but is actually incorrect, fabricated, or unsupported by facts. Hallucinations can happen when the model predicts text based on patterns rather than verified knowledge.

Example

Suppose a user asks:

"Who invented the smartphone in 1850?"

An AI might incorrectly respond with a specific person's name and detailed explanation, even though smartphones did not exist in 1850. The answer sounds convincing but is factually wrong. This

